Multiclass Prediction of Crop Yield Using Machine Learning Models

Rutendo Nesara

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1 Introduction

Crop yield prediction plays a critical role in modern agriculture by supporting decision-making, improving productivity, and optimizing the allocation of limited resources. In this project, the goal is to predict the crop yield performance by classifying it into three categories: Low, Medium, and High. Rather than modeling continuous yield values, the problem is formulated as a multiclass classification task to improve interpretability and provide actionable insights for farmers and agricultural planners. The target variable, `Yield_Category`, was created from the original continuous yield variable using quartile-based binning, resulting in three balanced classes. This ensures that each yield category is equally represented during model training and evaluation, allowing fair comparison between models. The dataset used in this study was obtained from Kaggle and contains approximately 10,000 agricultural observations. Each observation represents a single crop instance and includes a combination of soil properties, environmental conditions, and farming practices. The features consist of both numerical variables (such as nitrogen, phosphorus, potassium, rainfall, temperature, and soil moisture) and categorical variables (such as soil type, crop type, region, and irrigation method). These variables capture key factors known to influence agricultural productivity. This project aims to identify the most effective approach for predicting yield categories and to provide insights into the underlying factors that drive crop productivity.

2 Exploratory Data Analysis

The dataset contains approximately 10,000 observations with both numerical and categorical features. The target variable is balanced across three categories. No significant missing values were found. Key features such as rainfall and soil moisture show strong variation across yield categories, suggesting high predictive importance.

2.1 Feature Insights from Visualizations

Two visualizations were used to highlight the most informative relationships between features and yield categories: (1) boxplots of key variables and (2) a comparison of average feature values across categories.

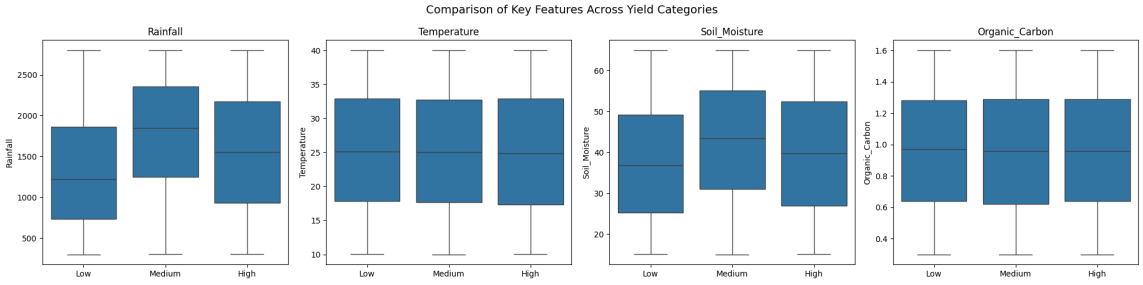


Figure 1: Distribution of key features across yield categories

The boxplots in Figure 1 show that rainfall and soil moisture exhibit the strongest separation across yield categories. Medium and high yield observations consistently occur under higher rainfall and moisture conditions, while Low yield observations are associated with reduced water availability. This pattern suggests that water-related variables are primary drivers of yield performance and are likely to have high predictive importance in the models..

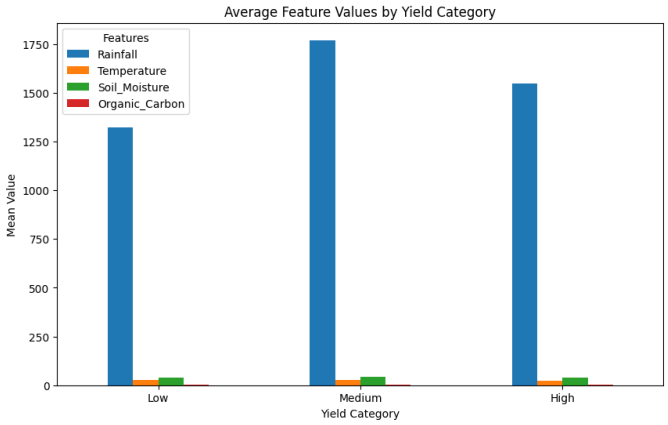


Figure 2: Average feature values by yield category

The average feature comparison in Figure 2 reinforces these findings. Rainfall and soil moisture show clear increases from Low to Medium yield categories, with Medium yield consistently exhibiting the highest mean values. In contrast, temperature and organic carbon display relatively small differences across categories, indicating weaker individual influence on yield outcomes. These patterns suggest that while some variables have a direct and substantial impact on yield, others may contribute through subtle or nonlinear interactions.

Overall, the exploratory analysis indicates that the dataset contains meaningful structure, with water-related features emerging as the most influential predictors. This supports the use of both linear models, which capture dominant trends, and nonlinear models, which may better capture complex interactions among environmental and soil variables.

3 Data Preparation and Preprocessing

Several data engineering steps were performed to prepare the dataset for modeling. The continuous crop yield variable was transformed into a categorical variable, *Yield_Category*, using quantile-based binning to create three classes: Low, Medium, and High. The original yield variable was removed from the feature set to prevent data leakage. The dataset was split into three subsets: 80% training, 10% validation, and 10% test. The same split was used across all models. The validation set was used for hyperparameter tuning and neural network training, while the test set was reserved for final evaluation.

A preprocessing pipeline was implemented using a *ColumnTransformer* to handle different feature types. For numerical features, missing values were imputed using the median, and features were standardized for models sensitive to scale (e.g., Logistic Regression and Neural Networks). For categorical features, missing values were imputed using the most frequent category and encoded using one-hot encoding to avoid introducing ordinal relationships.

No text or time-based features were present, so TF-IDF and cyclic feature engineering were not required. No irrelevant identifiers were included, so no additional feature exclusion was necessary. The target variable required no further transformation beyond binning. Since the dataset was balanced due to quantile-based binning, no resampling techniques were applied.

4 Model Training and Results

Logistic Regression achieved the best performance with the highest test F1 score (0.89) and low variance, indicating strong and stable predictions across cross-validation folds. Decision Tree showed the lowest performance, likely due to overfitting. The Support Vector Classifier (SVC) achieved strong performance with a test F1 score of 0.88, comparable to ensemble methods such as Random Forest and Gradient Boosting. This indicates that SVC is effective in capturing complex decision boundaries, but does not outperform Logistic Regression.

Random Forest and Gradient Boosting also performed well, achieving high F1 scores by capturing nonlinear relationships and improving generalization through ensemble learning. In particular, Gradient Boosting improves performance iteratively by correcting errors from previous models, allowing it to model complex patterns in the data.The Neural Network showed competitive performance, with training curves indicating effective learning and only mild overfitting controlled by dropout and early stopping.

Overall, while several models achieved similar performance, Logistic Regression remains the best-performing and simplest model, suggesting that the dataset is well-suited for linear decision boundaries.Hyperparameter tuning was performed using grid search for Logistic Regression, Decision Tree, Random Forest, and SVC, while Gradient Boosting and the Neural Network were tuned using a validation-based approach with early stopping. The low standard deviation of cross-validation scores further indicates consistent model performance across different data splits.

Table 1: Model Comparison

Model	Params	CV	Std	Val F1	Test F1	Acc
Logistic Reg.	C=0.01	0.87	0.007	0.87	0.89	0.84
Decision Tree	Tuned	N/A	N/A	0.82	0.84	0.84
SVC	C, gamma, kernel	—	—	0.88	0.88	0.88
Random Forest	Tuned	N/A	N/A	0.86	0.88	0.88
Grad. Boosting	n=100, lr=0.05, d=3	—	—	0.87	0.88	0.88
Neural Net	128-64 + dropout	N/A	N/A	0.87	0.88	0.88

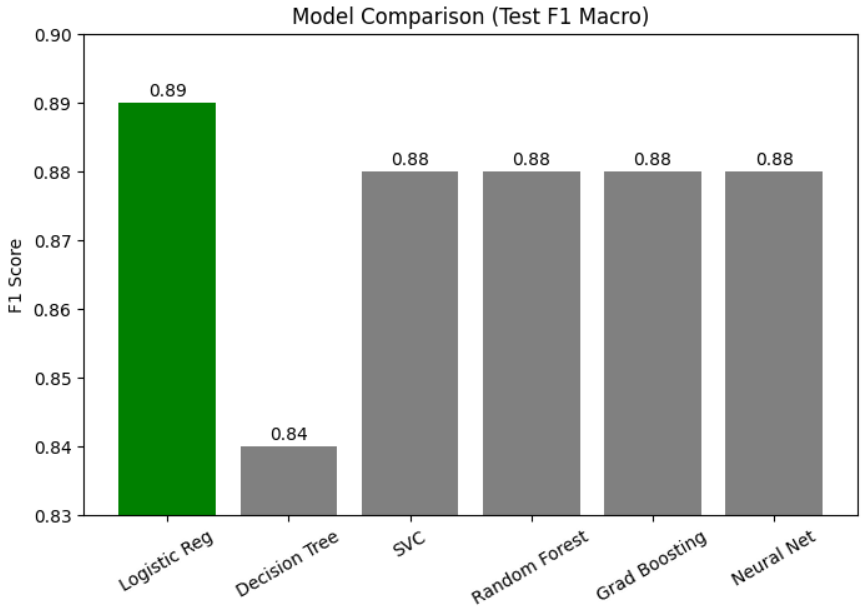


Figure 3: Comparison of test F1 scores across all models. Logistic Regression achieves the highest performance, although differences between models are small.

5 Conclusion and Next Steps

In this study, multiple machine learning models were developed to predict crop yield categories using environmental and agricultural features. Model performance was evaluated using the F1 Macro score, which ensures balanced evaluation across all classes.

Among all models, Logistic Regression achieved the best performance, with the highest test F1 score (0.89) and low variability across cross-validation folds. This indicates that the model is both accurate and stable. Although more complex models such as Support Vector Classifier, Random Forest, Gradient Boosting, and Neural Network achieved comparable performance, they did not significantly outperform Logistic Regression.

These results suggest that the dataset contains patterns that can be effectively captured using a linear model, and increasing model complexity does not necessarily lead to improved performance. Decision Tree showed the lowest performance, likely due to overfitting, highlighting the advantage of ensemble methods.

Future Scope

Future improvements could include:

- Exploring advanced ensemble methods such as XGBoost or LightGBM.
- Performing additional feature engineering to capture interactions between environmental and soil variables.
- Incorporating external data sources such as geographic or seasonal information.
- Evaluating model performance on datasets from different regions to assess generalizability.